



Last updated: 30th April 2026
Due to our policy on ongoing development, dimensions and specifications may change without notice.

Platform Load Cell | Single Point Load Cell | 0-250g to 0- 40kg | OBUG

Outstanding Accuracy, Lightweight and Low Cost, Low Range Compression Measuring

- **Lead Time:** 2 weeks
- **Buy online:** <https://appmeas.co.uk/shop/load-cells/obug/>



AT A GLANCE

- Capacities: 0-250g up to 0-40kg
- Output: 1.5-2mV/V
- Aluminium Construction
- High Accuracy: $<\pm 0.03\%/RC$
- Customised Versions Available

- **Designed for Low Range Measurement Applications**
- **Ideal in Weighing Scales**
- **Suitable for Industrial Environments**
- **Ideal for Retrofitting**
- **Lightweight & Low Cost**
- **High Accuracy Guaranteed**
- **Environmental Protection: IP65 ($\leq 2\text{kg}$ / IP67 ($\geq 3\text{kg}$))**
- **Optional ATEX Certification ($\geq 3\text{kg}$): EEx ia IIC T4**

DESCRIPTION

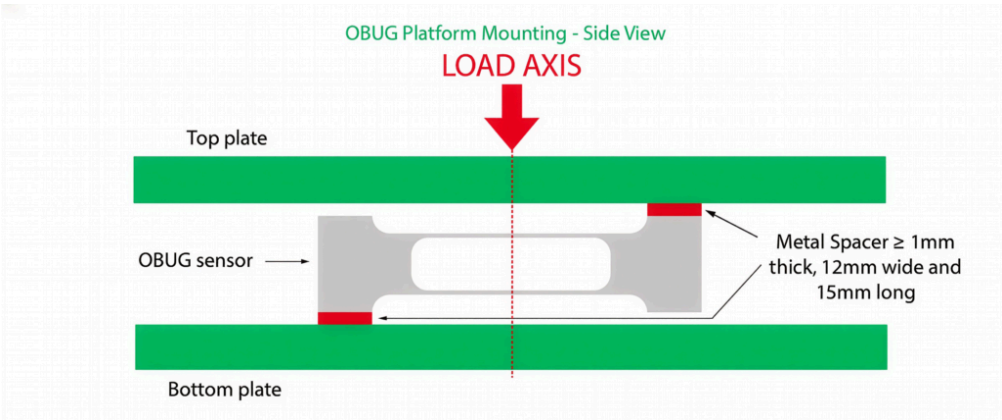


Applied Measurements low profile single point platform load cell OBUG is designed for use in weighing scales and a wide selection of low range force measurement applications such as manufacturing, testing, research and development. It is primarily designed for compression measurements, but it can also measure tension loads, or it can be used bi-directionally.

The simple hollowed out, dual cantilever design of the single point platform load cell makes it insensitive to bending moments and also offers extremely high accuracy and stability, while aluminium construction ensures light weight and low cost.

Applied Measurements can provide customised versions of the single point platform load cell with dimensions and fixings to allow direct retro-fitment in place of obsolete products or to suit specific applications are available on request.

RECOMMENDED MOUNTING



[single_testimonial id='5507']

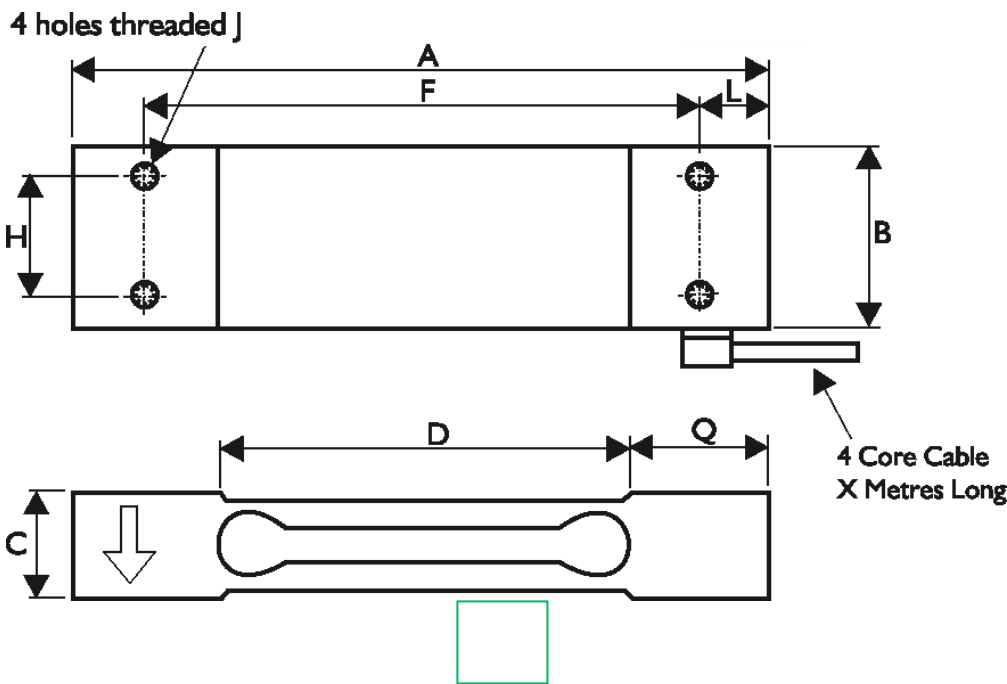
TECHNICAL SPECIFICATIONS

CHARACTERISTICS	OBUG	OBUG	OBUC	UNITS
Rated Capacity (RC)	0-0.25, 0-0.5, 0-1, 0-2	0-3, 0-5, 0-10, 0-15, 0-20, 0-30, 0-40	0-10, 0-15, 0-20, 0-30, 0-50, 0-100, 0-200	kgf
Operating Modes	Compression only			
Sensitivity (RO)	1.5 nominal	2±10%	2±10%	mV/V
Maximum Platform Dimensions	200 x 200	350 x 350	400 x 400	mm
Zero Balance/Offset	<1	<1	<10	±%/Rated Output
Creep (30 mins)	<0.08	<0.03		±%/Rated Capacity
Non-Linearity	<0.03			±%/Rated Capacity
Hysteresis	<0.02			%/Rated Capacity
Repeatability	<0.02			±%/Rated Capacity
Temperature Effect on Zero	<0.005	<0.002	<0.006	±%/Rated Capacity/°C
Temperature Effect on Sensitivity	<0.015	<0.002	<0.015	±%/Applied Load/°C
Effect of Eccentricity	<0.015	<0.006		±%/Rated Capacity/cm
Input Resistance	400 nominal			Ohms
Output Resistance	350 nominal			Ohms
Insulation Resistance	>2000 @ 50Vdc			Megohms
Excitation Voltage	10 recommended (2-15 acceptable)			Volts AC or DC
Operating Temperature Range	-20 to +70			°C
Compensated Temperature Range	-10 to +40			°C
Storage Temperature Range	-20 to +70			°C
Safe Overload	150			% of Rated Capacity



CHARACTERISTICS	OBUG		OBUG	OBUC	UNITS
Ultimate Overload			300		% of Rated Capacity
Deflection @ Rated Capacity			<0.5 nominal		mm
Resonant Frequency			Consult Sales Team		
IP Rating (Environmental Protection)			IP65: 2kg and below IP67: 3kg and above		
Weight (excluding cable)			See dimension table		
Fatigue Life			Consult Sales		
Cable Length (as standard)	0.5		0.4	3	metres
Cable Type	4 core screened, PVC sheath		4 core screened, PVC sheath,	6 core screened, PVC sheath,	
Construction			Aluminium Alloy		
Resolution			1 part in 250,000 (with appropriate instrumentation)		

Product Dimensions



Capacity (kg)	A	B	C	D	F	G	H	L	Q	V	X	J	Weight (kg)
0.25, 0.5, 1, 2	70	15	22	46	58	-	7	6	12	-	0.5	M3x0.5 - 22 deep	0.07
3, 5, 10, 15, 20, 30	130	25.4	22	80	106	-	15	12	25	-	0.4	M6x1 - 22 deep	0.20
40	130	30	22	80	106	-	15	12	25	-	0.4	M6x1 - 22 deep	0.24

All dimensions are in mm

Wiring Details (Capacities of 2kg and Lower)

Wire	Designation
Red	+ve excitation
Blue	-ve excitation
Green	+ve signal
Yellow	-ve signal
Screen	To ground – not connected to load cell body



Wiring Details (Capacities of 3kg to 40kg)

Wire	Designation
Red	+ve excitation
Black	-ve excitation
Green	+ve signal
White	-ve signal
Screen	To ground – not connected to load cell body

ORDERING CODES & OPTIONS

Core Product	Capacity (inc Engineering Units)	Cable Length (m)	Specials Code	Example Result
OBUG	0.25kg	0.5	000	OBUG-0.25kg-0.5-000
OBUG	0.5kg	0.5	000	OBUG-0.5kg-0.5-000
OBUG	1kg	0.5	000	OBUG-1kg-0.5-000
OBUG	2kg	0.5	000	OBUG-2kg-0.5-000
OBUG	3kg	0.4	000	OBUG-3kg-0.4-000
OBUG	5kg	0.4	000	OBUG-5kg-0.4-000
OBUG	10kg	0.4	000	OBUG-10kg-0.4-000
OBUG	15kg	0.4	000	OBUG-15kg-0.4-000
OBUG	20kg	0.4	000	OBUG-20kg-0.4-000
OBUG	30kg	0.4	000	OBUG-30kg-0.4-000
OBUG	40kg	0.4	000	OBUG-40kg-0.4-000

PUBLISHED SENSOR APPLICATION ARTICLES

Below is a published sensor application paper that shows you how the OBUG platform load cell could be used in a specific application. See our published sensor application articles page (<https://appmeas.co.uk/resources/sensor-application-papers/>) for many more.

An experimental investigation of tsunami forces on coastal structures (<https://discovery.ucl.ac.uk/id/eprint/1522021/>)

Lloyd, TO; (2016) An experimental investigation of tsunami forces on coastal structures. Doctoral thesis, UCL (University College London). Available as open access.

CASE STUDIES



How to Improve Aerodynamics with our Single Point Load Cells and Digital Indicators

The Scale Wind Tunnel uses two single point load cells and Intuitive2-L digital indicators enabling students to test the aerodynamics of scale models. Read how they did it.

Read more... (<https://appmeas.co.uk/blog/improve-aerodynamics-single-point-load-cells-digital-indicators/>)



Single Point Load Cells Help Farmers Balance the Scales

The Challenge – Design a weighing system to accurately measure the weight of birds maximising poultry farming profitability. Poultry farmers need to accurately weigh the birds to ensure they meet the correct specification and adjust the bird feed mix accordingly.



Read more... (<https://appmeas.co.uk/blog/single-point-load-cells-help-farmers-balance-scales/>)

RELATED PRODUCTS

Buy Online



**Load Cell Digital Indicator | High Accuracy
| Intuitive4-L**
Highly configurable, modular construction
From **£364**

Buy Online



**Load Cell Amplifier | Load Cell Signal
Conditioner | SGA**
Inputs from 0.06mV/V
From **£235**

Buy Online



USB Load Cell Interface | DSCUSB
100 Readings/Second max.
From **£335**

Buy Online



**Miniature Load Cell Amplifier | In-Cell
Amplifier | ICA**
From **£85**

Buy Online



**Handheld Load Cell Indicator | Digital
Display | TR150**
From **£495**



Large Digit Load Cell Display
From **£503**



T24 Wireless Telemetry

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